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Introduction

In this report, we price a down-and-out barrier call option using a stochastic differential equation (SDE) to model the stock price dynamics:

$$dS_t = rS_t dt + \sigma S_t dW_t. \quad (1)$$

where S_t represents the stock price at time t , r is the risk-free interest rate, σ is the stock's volatility, and dW_t is the increment of a standard Brownian motion.

To solve the SDE (1), we employ the Milstein scheme and use a Monte Carlo simulation with $M = 10,000$ paths to estimate the expected payoff and standard deviation of the option. Our objective is to ensure that the 95% confidence interval of the option price estimate is within \$0.01. Achieving this level of precision requires a large number of Monte Carlo paths, which is computationally expensive. To mitigate this, we implement variance reduction techniques to decrease the required number of paths while maintaining accuracy.

We compare the computational efficiency and simulation requirements across different methodologies. The input parameters for our models are as follows: initial stock price $S_0 = \$100$, volatility $\sigma = 0.1$, interest rate $r = 0.01$, maturity $T = 1.5$ years, strike price $K = \$90$, barrier level $B = \$61$, and number of time steps $N = 1000$.

Pricing the Option

In this section, we implement the baseline case without variance reduction techniques. This involves computing the price of a down-and-out barrier call option using the Milstein scheme for the numerical solution of the SDE. The Milstein scheme is given by:

$$S_{i+1} = S_i + \mu(S_i, t_i)\Delta t + \sigma(S_i, t_i)dW_i + \frac{1}{2}\sigma(S_i, t_i)\sigma'(S_i, t_i)(dW_i^2 - \Delta t). \quad (2)$$

where $\mu(S_i, t_i)$ is the drift coefficient which corresponds to rS_t in our SDE and $\sigma(S_i, t_i)$ is the diffusion coefficient which corresponds to σS_t in our SDE.

Using this discretisation, we implement **Algorithm 1**, which outlines the Monte Carlo simulation procedure used in Python to estimate the option price. The algorithm simulates multiple stock price paths, applies the barrier condition, computes the expected payoff, and constructs a 95% confidence interval for the estimated option price.

Algorithm 1 Compute the price of a down-and-out barrier call option using the Milstein scheme

Input: $S_0, K, B, r, \sigma, T, N, M$

Compute $\Delta t = T/N$

Initialise $S_{0,m} = S_0$ and $B_m = \text{False}$ for all $m = 1, \dots, M$

for $i = 1$ **to** N **do**

 Generate $Z_{i,m} \sim \mathcal{N}(0, 1)$ and set $dW_{i,m} = \sqrt{\Delta t} \cdot Z_{i,m}$

 Update stock prices (using the Milstein scheme):

$$S_{i,m} = S_{i-1,m} + rS_{i-1,m}\Delta t + \sigma S_{i-1,m}dW_{i,m} + 0.5\sigma^2 S_{i-1,m}(dW_{i,m}^2 - \Delta t)$$

 Update barrier condition:

$$B_m = \begin{cases} \text{True,} & \text{if } S_{i,m} < B \text{ for some } i = 1, \dots, N \\ \text{False,} & \text{otherwise} \end{cases}$$

end for

Compute payoffs:

$$h(S_{N,m}) = \begin{cases} 0, & \text{if } B_m = \text{True (i.e., the barrier was breached at some point)} \\ \max(S_{N,m} - K, 0), & \text{otherwise} \end{cases}$$

Compute Monte Carlo estimates (expected payoff and standard deviation of the option):

$$a_M = e^{-rT} \cdot \mathbb{E}[h(S_{N,m})], \quad b_M = e^{-rT} \cdot \sqrt{\text{Var}(h(S_{N,m}))}$$

$$\text{Compute 95\% Confidence Interval: } CI = \left[a_M - 1.96 \frac{b_M}{\sqrt{M}}, \quad a_M + 1.96 \frac{b_M}{\sqrt{M}} \right]$$

Output: a_M, b_M, CI

We then calculate the number of paths required in a Monte Carlo simulation to achieve a confidence interval below \$0.01 as follows:

$$M^* = \left\lceil \left(\frac{2 \cdot 1.96 \cdot b_M}{0.01} \right)^2 \right\rceil. \quad (3)$$

where $\lceil \cdot \rceil$ is the ceiling of our computation, as the number of Monte Carlo paths can only take an integer value.

We implement our algorithm with our specific parameters. We then calculate the computational time for the case with $M = 10,000$ paths. We assume the computational time increases linearly with M , meaning we can estimate the computational time to achieve a confidence interval less than \$0.01 as follows:

$$t^* = \left(\frac{M^*}{M} \right) \cdot t. \quad (4)$$

where t is the computational time for a Monte Carlo simulation with $M = 10,000$ paths.

Table 1 summarises the results of the base case, showing that a large number of Monte Carlo paths are required to achieve a 95% confidence interval below \$0.01, leading to significant computational time. Since increasing M is computationally inefficient, we instead implement variance reduction techniques to lower b_M and achieve the desired confidence interval more efficiently.

a_M (\$)	b_M (\$)	CI (\$)	t (seconds)	M^*	t^* (seconds)
12.4907	11.0165	[12.2748, 12.7066]	0.2870	18649312	535.21

Table 1: Milstein Scheme Results

Variance Reduction: Antithetic Variates

In this section, we aim to reduce the number of Monte Carlo simulations required to achieve a 95% confidence interval below \$0.01 by using antithetic variates. This variance reduction technique works by simulating an additional stock price path using negatively correlated Brownian increments. The general structure remains the same as in **Algorithm 1**, but instead of generating a single standard normal variable $Z_{i,m}$ per time step and per path, we now generate pairs of negatively correlated standard normal variables for each path:

$$Z_{i,m}, -Z_{i,m} \sim \mathcal{N}(0, 1). \quad (5)$$

This ensures that the generated stock price paths remain negatively correlated at each time step for each path m . The corresponding Brownian increments for the two paths are:

$$dW_{i,m} = \sqrt{\Delta t} \cdot Z_{i,m}, \quad -dW_{i,m} = -\sqrt{\Delta t} \cdot Z_{i,m}. \quad (6)$$

Using these increments, we simulate two stock price paths in parallel based on the Milstein scheme:

$$S_{i+1,m}^+ = S_{i,m}^+ + rS_{i,m}^+ \Delta t + \sigma S_{i,m}^+ dW_{i,m} + \frac{1}{2} \sigma^2 S_{i,m}^+ (dW_{i,m}^2 - \Delta t), \quad (7)$$

$$S_{i+1,m}^- = S_{i,m}^- + rS_{i,m}^- \Delta t + \sigma S_{i,m}^- (-dW_{i,m}) + \frac{1}{2} \sigma^2 S_{i,m}^- ((-dW_{i,m})^2 - \Delta t), \quad (8)$$

where $S_{i,m}^+$ and $S_{i,m}^-$ represent the stock prices for the standard and antithetic paths at time step i , respectively, and for each simulation path m .

The barrier condition is applied separately to each path, meaning we now track two barrier conditions, one for each trajectory. The payoffs are computed as in **Algorithm 1**, and the final option price and standard deviation are given by:

$$a_M = e^{-rT} \cdot \mathbb{E} \left[\frac{1}{2} \left(h(S_{N,m}^+) + h(S_{N,m}^-) \right) \right], b_M = e^{-rT} \cdot \sqrt{\text{Var} \left(\frac{1}{2} \left(h(S_{N,m}^+) + h(S_{N,m}^-) \right) \right)}. \quad (9)$$

Before analysing the results, we validate the effectiveness of the antithetic variates implementation by computing the correlation of the final stock prices and option payoffs for both paths. As shown in Table 2, the correlation is strongly negative in both cases, confirming the successful implementation of antithetic variates.

Final Stock Price Correlation	Final Option Price Correlation
-0.985190	-0.855489

Table 2: Correlation of the Final Stock Price and Option Price of the Two Sequences

Table 3 summarises the results of the Milstein scheme with antithetic variates. The standard deviation of the option price is significantly reduced compared to the base case, requiring far fewer Monte Carlo paths to achieve a 95% confidence interval below \$0.01. The speedup factor quantifies the efficiency gain from this reduction. Clearly, antithetic variates enable us to reach the required precision far more efficiently than the base case.

a_M (\$)	b_M (\$)	CI (\$)	t (seconds)	M^*	t^* (seconds)	Speedup Factor
12.3413	2.9343	[12.2838, 12.3988]	0.4093	1323058	54.15	9.8831×

Table 3: Milstein Scheme with Antithetic Variates Results

Variance Reduction: Fine-Tuned Control Variates

In this section, we aim to reduce the number of Monte Carlo paths required to achieve a 95% confidence interval below \$0.01 by using fine-tuned control variates. This method follows a two-step process. First, we estimate the covariance between the option payoff $h(S_{N,m})$ and the final stock price $S_{N,m}$, as well as the variance of $S_{N,m}$, using a Monte Carlo simulation with $M = 10,000$ paths, as this is a sufficiently small number of paths. The optimal control variate parameter θ_{opt} is given by:

$$\theta_{\text{opt}} = \frac{\text{Cov}(h(S_{N,m}), S_{N,m})}{\text{Var}(S_{N,m})} \quad (10)$$

which minimises the variance of the adjusted estimator $Y_{\theta,m}$:

$$\text{Var}(Y_{\theta,m}) = \text{Var}(h(S_{N,m})) - 2\theta \text{Cov}(h(S_{N,m}), S_{N,m}) + \theta^2 \text{Var}(S_{N,m}). \quad (11)$$

Since this is a quadratic function in θ , it attains its minimum at θ_{opt} .

Next, a full Monte Carlo simulation with $M = 10,000$ paths is performed using the control variate approach. The adjusted estimator is defined as:

$$Y_{\theta_{\text{opt}},m} = h(S_{N,m}) + \theta_{\text{opt}} (\mathbb{E}[S_{N,m}] - S_{N,m}). \quad (12)$$

where $\mathbb{E}[S_{N,m}] = S_0 e^{rT}$.

Since $\mathbb{E}[Y_{\theta_{\text{opt}},m}] = \mathbb{E}[h(S_{N,m})]$, this ensures the estimator remains unbiased while reducing variance.

The stock price evolution follows the Milstein discretisation seen in **Algorithm 1**, with the barrier condition enforced and payoffs computed in the same manner. The final price of the option and its standard deviation are then calculated as:

$$a_M = e^{-rT} \cdot \mathbb{E}[Y_{\theta_{\text{opt}},m}], \quad b_M = e^{-rT} \cdot \sqrt{\text{Var}(Y_{\theta_{\text{opt}},m})}. \quad (13)$$

To validate the effectiveness of our implementation of fine-tuned control variates, we plot $\text{Var}(Y_{\theta,m})$ as a function of θ , ensuring that we have found the minimum variance and that it is lower than in the base case, i.e., $\text{Var}(Y_{\theta_{\text{opt}},m}) < \text{Var}(h(S_{N,m}))$. As shown in Figure 1, this is indeed the case, confirming the correctness of our implementation.

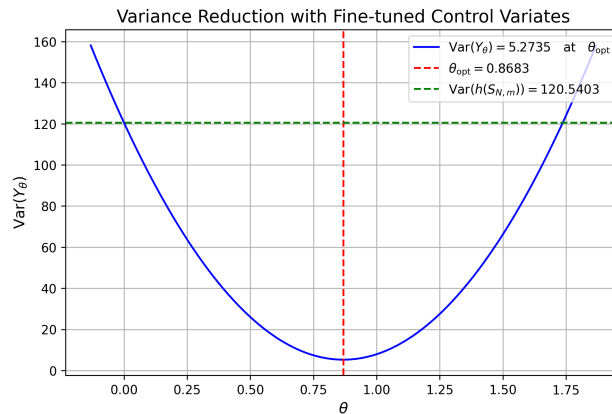


Figure 1: Variance Reduction with Fine-Tuned Control Variates

Table 4 summarises the results of the Milstein scheme with fine-tuned control variates. The standard deviation of the option price is significantly reduced compared to the base case, requiring far fewer Monte Carlo paths to achieve a 95% confidence interval below \$0.01. The speedup factor quantifies the efficiency gain from this reduction. Clearly, fine-tuned control variates enable us to reach the required precision far more efficiently than the base case. Additionally, we note that fine-tuned control variates provide a greater variance reduction and speedup factor than antithetic variates, reinforcing their effectiveness as a variance reduction technique.

a_M (\$)	b_M (\$)	CI (\$)	t (seconds)	M^*	t^* (seconds)	Speedup Factor
12.3109	2.2622	[12.2665, 12.3552]	0.2850	786392	22.41	23.8813×

Table 4: Milstein Scheme with Fine-Tuned Control Variates Results

Variance Reduction: Antithetic and Fine-Tuned Control Variates

In this section, we combine the two variance reduction techniques used in the previous sections to obtain a more efficient Monte Carlo estimation of the down-and-out barrier call option price. The order in which these techniques are applied affects the variance of the Monte Carlo estimator (the option price), in turn influencing the number of Monte Carlo samples M required to achieve a 95% confidence interval below \$0.01. We analyse both cases theoretically and in practice.

Fine-Tuned Control Variates $\rightarrow$ Antithetic Variates First, we apply the fine-tuned control variate transformation separately to each path, as seen in Equation (12). This adjustment removes a significant amount of variance from each path independently before applying the antithetic variate method. By substituting Equation (10) into Equation (11), it can be shown that:

$$\text{Var}(Y_{\theta_{\text{opt}},m}) = \text{Var}(h(S_{N,m})) (1 - \rho^2) \quad (14)$$

where ρ^2 is the squared correlation coefficient between $h(S_{N,m})$ and $S_{N,m}$.

From this, it is clear that the stronger the correlation, the greater the variance reduction achieved. Since barrier events may differ between the original and antithetic paths, we estimate separate optimal parameters θ_{opt}^+ and θ_{opt}^- for each path individually:

$$\begin{aligned} Y_{\theta_{\text{opt}},m}^+ &= h(S_{N,m}^+) + \theta_{\text{opt}}^+ \left(\mathbb{E}[S_{N,m}] - S_{N,m}^+ \right), \\ Y_{\theta_{\text{opt}},m}^- &= h(S_{N,m}^-) + \theta_{\text{opt}}^- \left(\mathbb{E}[S_{N,m}] - S_{N,m}^- \right). \end{aligned} \quad (15)$$

Next, we introduce the antithetic variates method by averaging these negatively correlated adjusted payoffs, ensuring further variance reduction:

$$a_M = e^{-rT} \cdot \mathbb{E} \left[\frac{1}{2} \left(Y_{\theta_{\text{opt}},m}^+ + Y_{\theta_{\text{opt}},m}^- \right) \right]. \quad (16)$$

Since control variates have already removed unstructured variance (noise), the remaining variance exhibits a stronger correlation structure. We calculate the standard deviation of the payoff as follows:

$$b_M = e^{-rT} \cdot \sqrt{\text{Var} \left(\frac{1}{2} \left(Y_{\theta_{\text{opt}},m}^+ + Y_{\theta_{\text{opt}},m}^- \right) \right)}. \quad (17)$$

With the control variate transformation already reducing variance, averaging negatively correlated antithetic paths further enhances variance reduction. Since the variance of an average of two correlated estimators is lower than that of each individual estimator, applying the antithetic variates method after control variates ensures minimal remaining variance. This compounding effect leads to greater overall variance reduction than using either method alone.

Table 5 summarises the results of this approach. Combining fine-tuned control variates with antithetic variates significantly reduces the number of Monte Carlo paths required to achieve a 95% confidence interval below \$0.01, requiring the fewest paths among all tested methods. Additionally, this combined method achieves the lowest standard deviation for the option price. However, the extra covariance and variance calculations, along with simultaneously simulating and adjusting both standard and antithetic paths, introduce additional computational complexity. This explains why the speedup factor, although substantial compared to the base method, is somewhat smaller than when using fine-tuned control variates alone. Nonetheless, the substantial reduction in the required number of paths results in a clear overall efficiency improvement.

a_M (\$)	b_M (\$)	CI (\$)	t (seconds)	M^*	t^* (seconds)	Speedup Factor
12.3487	2.0653	[12.3083, 12.3892]	0.3713	655463	24.34	21.9893×

Table 5: Milstein Scheme with Fine-Tuned Control Variates $\rightarrow$ Antithetic Variates Results

Antithetic Variates $\rightarrow$ Fine-Tuned Control Variates In this case, we first generate two correlated paths S^+ and S^- as seen in Equation (7) and Equation (8). However, since control variates work best when applied to high-variance data, applying them after antithetic variates limits their effectiveness. Applying the control variate method post-antithetic variates gives:

$$Y_{\theta_{\text{opt}},m} = \frac{1}{2} \left(h(S_{N,m}^+) + h(S_{N,m}^-) \right) + \theta_{\text{opt}} \left(\mathbb{E}[S_{N,m}] - S_{N,m}^+ \right). \quad (18)$$

We use the original stock price path $S_{N,m}^+$ for calculation of θ_{opt} and in this equation for the stock price part. We don't use an average of the two paths for this part, as this artificially removes the variance related to barrier events, thus causing an inflated, unrealistic variance reduction.

We then calculate the Monte Carlo estimates as:

$$a_M = e^{-rT} \cdot \mathbb{E}[Y_{\theta_{\text{opt}},m}], \quad b_M^2 = e^{-rT} \cdot \sqrt{\text{Var}(Y_{\theta_{\text{opt}},m})}. \quad (19)$$

As can be seen in Equation (14) reduction in variance is directly proportional to the initial variance. Thus, control variates are more effective with high-variance data. Since $\text{Var}(h(S_{N,m}))$ is already reduced by the antithetic variates, the impact of control variates is diminished.

Table 6 summarises the results of this approach. The findings support the theory that applying fine-tuned control variates first is the most efficient choice. In contrast, applying control variates after antithetic variates provides no significant efficiency gains beyond using antithetic variates alone and results in a lower speedup factor and standard deviation than fine-tuned control variates alone. These results suggest that, in this order, combining the two methods is inefficient, as the variance reduction achieved by antithetic variates diminishes the additional benefit of control variates, making one of the techniques redundant.

a_M (\$)	b_M (\$)	CI (\$)	t (seconds)	M^*	t^* (seconds)	Speedup Factor
12.3365	2.9621	[12.2784, 12.3945]	0.3713	1348226	50.06	10.6908×

Table 6: Milstein Scheme with Antithetic Variates $\rightarrow$ Fine-Tuned Control Variates Results

Results and Discussion

Overall, we conclude that among the implemented methods, the most efficient approach for estimating the price of a down-and-out barrier call option with a 95% confidence interval below \$0.01 is to first apply fine-tuned control variates, followed by antithetic variates to the solution of the SDE approximated by the Milstein scheme. This method resulted in the lowest standard deviation of the option price and required the fewest Monte Carlo samples. However, the highest speedup factor was achieved when using fine-tuned control variates alone. Figure 2 highlights our findings, illustrating the trade-off: while control variates alone offer the fastest computation time, combining them with antithetic variates minimises the standard deviation and the number of required Monte Carlo samples. This demonstrates that the optimal choice depends on whether the primary goal is computational speed or variance reduction.

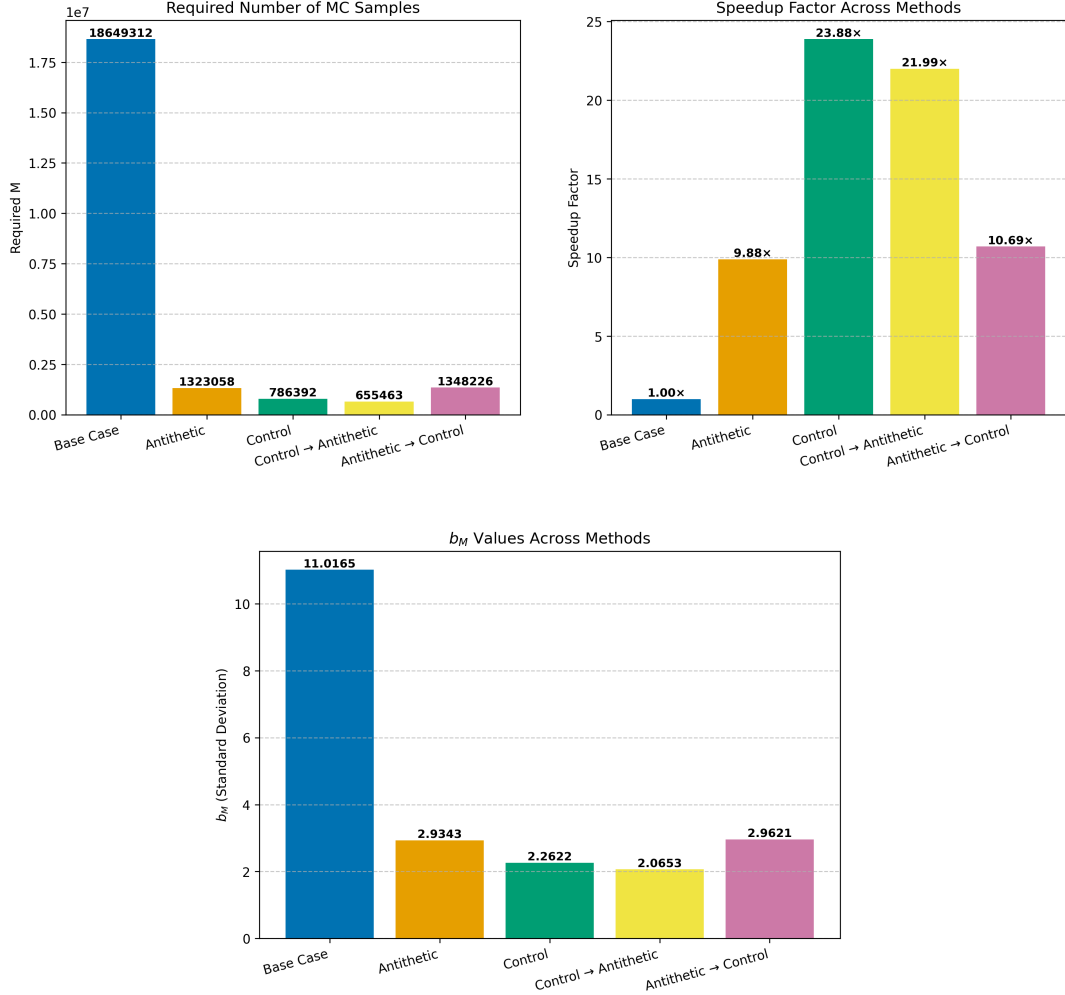


Figure 2: Comparing the Different Methods

Table 7 presents the results of the fine-tuned control variates followed by the antithetic variates method, using the minimum number of Monte Carlo samples required to achieve a 95% confidence interval below \$0.01. In theory, the option price estimate should remain unchanged from the case where $M = 10,000$, as it is an unbiased estimator. However, minor fluctuations occur in practice due to sampling variability.

a_M (\$)	b_M (\$)	CI (\$)	t (seconds)
12.3333	2.0315	[12.3284, 12.3383]	95.45052033299999

Table 7: Option Price Estimate with a 95% CI < \$0.01 for Fine-Tuned Control Variates then Antithetic Variates